

### 5.3.16 Extended interrupt and event controller input (EXTI) characteristics

The pulse on the interrupt input must have a minimal length in order to guarantee that it is detected by the event controller.

**Table 51. EXTI input characteristics**

Specified by design and not tested in production.

| Symbol | Parameter                        | Conditions | Min | Typ | Max | Unit |
|--------|----------------------------------|------------|-----|-----|-----|------|
| PLEC   | Pulse length to event controller | -          | 20  | -   | -   | ns   |

### 5.3.17 XSPI interface characteristics

Unless otherwise specified, the parameters given in the following table are derived from tests performed under the ambient temperature ,  $f_{HCLK}$  frequency, and  $V_{DD}$  supply voltage conditions summarized in [Table 16. General operating conditions](#) , with the following configuration:

- Output speed is set to  $OSPEEDR[1:0] = 11$
- Measurement points are done at CMOS levels:  $0.5 V_{DD}$
- IO compensation cell activated

**Table 52. XSPI characteristics in SRD mode**

Evaluated by characterization - Not tested in production.

All values apply to Octal- and Quad-SPI mode.

Delay block bypassed.

| Symbol        | Parameter                                      | Conditions                                             | Min                                    | Typ | Max                              | Unit |
|---------------|------------------------------------------------|--------------------------------------------------------|----------------------------------------|-----|----------------------------------|------|
| $f_{(CLK)}$   | Clock frequency                                | $2.7 V < V_{DD} < 3.6 V$<br>$C_L = 15 pF$              | -                                      | -   | 72                               | MHz  |
| $t_{w(CLKH)}$ | OCTOSPI clock high and low time, even division | PRESCALER[7:0] = $n$<br>$n = (0, 1, 3, 5, \dots, 255)$ | $t_{(CLK)}/2 - 0.5$                    | -   | $t_{(CLK)}/2 + 0.5$              | ns   |
| $t_{w(CLKL)}$ |                                                |                                                        | $t_{(CLK)}/2 - 0.5$                    | -   | $t_{(CLK)}/2 + 0.5$              |      |
| $t_{w(CLKH)}$ | OCTOSPI clock high and low time, odd division  | PRESCALER[7:0] = $n$<br>$n = (2, 4, 6, \dots, 254)$    | $(n/2) \times t_{(CLK)}/(n+1) - 0.5$   | -   | $(n/2) \times t_{(CLK)}/(n+1)$   |      |
| $t_{w(CLKL)}$ |                                                |                                                        | $(n/2+1) \times t_{(CLK)}/(n+1) - 0.5$ | -   | $(n/2+1) \times t_{(CLK)}/(n+1)$ |      |
| $t_{s(DQ)}$   | Data input setup time                          | -                                                      | 3                                      | -   | -                                |      |
| $t_{h(DQ)}$   | Data input hold time                           |                                                        | 1                                      | -   | -                                |      |
| $t_{v(OUT)}$  | Data output valid time                         |                                                        | -                                      | 0.5 | 1.5                              |      |
| $t_{h(OUT)}$  | Data output hold time                          |                                                        | 0                                      | -   | -                                |      |

**Table 53. XSPI characteristics in DTR mode (no DQS)**

Evaluated by characterization - Not tested in production.

All values apply to Octal- and Quad-SPI mode.

Delay block bypassed.

| Symbol        | Parameter                                     | Conditions                                             | Min                                    | Typ | Max                                    | Unit |
|---------------|-----------------------------------------------|--------------------------------------------------------|----------------------------------------|-----|----------------------------------------|------|
| $f_{(CLK)}$   | Clock frequency                               | $2.7 V < V_{DD} < 3.6 V$<br>$C_L = 15 pF$              | -                                      | -   | 72 <sup>(1)</sup>                      | MHz  |
| $t_{w(CLKL)}$ | OCTOSPI clock high and low time Even division | PRESCALER[7:0] = $n$<br>$n = (0, 1, 3, 5, \dots, 255)$ | $t_{(CLK)}/2 - 0.5$                    | -   | $t_{(CLK)}/2 + 0.5$                    | ns   |
| $t_{w(CLKH)}$ |                                               |                                                        | $t_{(CLK)}/2 - 0.5$                    | -   | $t_{(CLK)}/2 + 0.5$                    |      |
| $t_{w(CLKH)}$ | OCTOSPI clock high and low time Even division | PRESCALER[7:0] = $n$<br>$n = (2, 4, 6, \dots, 254)$    | $(n/2) \times t_{(CLK)}/(n+1) - 0.5$   | -   | $(n/2) \times t_{(CLK)}/(n+1) + 0.5$   |      |
| $t_{w(CLKL)}$ |                                               |                                                        | $(n/2+1) \times t_{(CLK)}/(n+1) - 0.5$ | -   | $(n/2+1) \times t_{(CLK)}/(n+1) + 0.5$ |      |